

Augmented Sample Size Calculation With Truncated Data

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Incomplete data, including truncated and censored observations, can lead to information loss and effect statistical inference. My work develops an effective sample size framework (ESS) to quantify information loss in incomplete data. An Expectation–Maximization (EM) algorithm is used to estimate the ESS, which is then incorporated into inference procedures by replacing the nominal sample size.

The effective sample size is defined as combining fully observed, truncated, and censored observations, the effective sample size can be written as

$$m_{\text{eff}} = \sum_{i \in O} 1 + \sum_{j \in T} w_j^{(T)} + \sum_{k \in C} w_k^{(C)}, \quad (1)$$

where O , T , and C denote the sets of fully observed, truncated, and censored observations, respectively. The weights $w_j^{(T)}$ and $w_k^{(C)}$ take values in $(0, 1]$ and quantify the relative information contribution of truncated and censored observations. They are computed from the corresponding truncation or censoring probabilities under the assumed distribution, with smaller weights assigned to observations containing greater information loss.

Simulation studies compare the proposed method with conventional approaches using confidence interval coverage, Type I error, power, bias, and mean squared error. Real-data applications are also considered. The results show that the proposed ESS effectively reflects information loss and generally provides better type I error and confidence coverage but poorer power than methods based only on the observed sample size.

References

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